Supplementary Information: Unveiling Mycobacterium marinum Cellular and Metabolic Remodeling Under Simulated Microgravity

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1 Supplementary Tables Overview

All empirical data tables are archived in the repository under `results_tables/` and available via Zenodo:

- **Table S1** (`Table_S1_raw_counts_matrix.tsv`): Complete raw transcript count matrix ($5,510$ genes$\times$ 9 samples) generated by `kallisto` v0.52.0 pseudoalignment of NextSeq 550 paired-end reads.
- **Table S2** (`Table_S2_normalized_log2cpm`$_{e}xpression.tsv) : Normalized \log_2($`CPM+1) matrix across 4,964 expressed genes.`
- `Table S3 (Table_S3_sample_metadata.tsv): Experimental metadata for NASA OSDR OSD-528.`
- `Table S4 (Table_S4_differential`$_{e}xpression_{3}dclinostat_{v}s_{1}g.tsv) : 351significantDEGsin3DClinostatv$
- `Table S5 (Table_S5_differential`$_{e}xpression_{r}pm2_{v}s_{1}g.tsv) : 738significantDEGsinRPM2.0vsStatic1g.$
- `Table S6 (Table_S6_differential`$_{e}xpression_{3}dclinostat_{v}s_{r}pm2.tsv) : 242DEGscomparingsimulationkin$
- `Table S7 (Table_S7_wgcna`$_{m}odule_{a}ssignments_{g}oslim.tsv) : WGCNAmoduleassignments, standardizedG$ `hub scores.`
- `Table S8 (Table_S8_wgcna`$_{m}odule_{e}igengenes.tsv) : ModuleEigengeneprofilesacrossall9samples.$

- Table S9 (Table_S9_wgcna$_{m}odule_{t}rait_{c}orrelations.tsv) : Pearsoncorrelationcoefficientsandasymptotic$p

- Table S10 (Table_S10$_{g}oslim_{p}athway_{e}nrichment.tsv) : GOSlimpathwayover-representationmetrics($k/r
$HochbergFDR).$

- Table S11 (Table_S11$_{f}ull_{g}ene_{o}ntology_{e}nrichment.tsv) : ComprehensiveGOenrichmenttableacrossBiolog$

- Table S12 (Table_S12$_{t}abpfn_{l}oocv_{p}redictions.tsv) : TabPFNleave-one-outcross-validationpredictions$

- Table S13 (Table_S13$_{t}abpfn_{f}eature_{i}mportance.tsv) : TabPFNpermutationfeatureimportancerankings$

- Table S14 (Table_S14$_{c}ellular_{m}etabolic_{m}odel_{r}eactions.tsv) : Reaction-levelGPRmappingacross32enzym$

- Table S15 (Table_S15$_{m}etabolic_{s}ubsystem_{p}erturbation_{i}ndex.tsv) : QuantitativeSubsystemPerturbationI$

- Table S16 (Table_S16$_{p}an_{m}icrobial_{o}sdr_{c}ompendium_{7}8studies.tsv) : Compendiumof78microbialspaceflig$

- Table S17 (Table_S17$_{c}ellular_{m}etabolic_{m}odel_{s}bml.json) : SBML-compatiblecomputationalmodelschema$

2 Supplementary Figures Overview

All supplementary figures are archived as vector PDFs and high-resolution 300 DPI PNGs in `supplementary_figures/`:

- **Figure S1**: Systems architecture, biological model, simulation hardware, and analytical framework.
- **Figure S2**: Transcriptomic PCA biplot (70.1% PC1 variance) and volcano plot of 351 DEGs.
- **Figure S3**: Gene count distribution across 5 GOSlim co-expression modules and module-trait heatmap.
- **Figure S4**: TabPFN tabular foundation model benchmark under LOOCV, confusion matrix, and feature importances.
- **Figure S5**: Enriched GOSlim functional pathways with calibrated FAIR Blue-White-Red color fill.
- **Figure S6**: Intramodular hub connectivity (k_{within} vs k_{total}) and inter-module regulatory interactome.
- **Figure S7**: Pan-microbial spaceflight meta-analysis landscape across 78 OSDR studies and concordance matrix.

- **Figure S8**: Hexagonal trajectory radar mapping phenotypic concordance across 6 physiological axes.
- **Figure S9**: Multi-compartment cellular cross-section and Subsystem Perturbation Index (SPI) ranking.

3 Read Quality Control Pseudoalignment Convergence

Pseudoalignment against *M. marinum* M strain (NC_010612.1) converged with high pseudoalignment efficiency across all 9 biological replicates (mean 1.21 million reads mapped per sample). Normalized counts were converted to log-CPM values using edgeR TMM library scaling. Zero synthetic data were utilized in any calculation.